components q_i, p_i, Q_i, P_i, respectively; we may think of Q_i and P_i as functions of $\boldsymbol{q}$ and $\boldsymbol{p}$, then

$$\mathrm{d}Q_i = \mathrm{d}q_i + hm_i^{-1}\mathrm{d}P_i, \tag{2.20}$$

$$\mathrm{d}P_i = \mathrm{d}p_i - h\sum_{j=1}^{\mathrm{N_c}} \frac{\partial^2 U}{\partial q_j \partial q_i}\mathrm{d}q_j. \tag{2.21}$$

Computing the 2-form,

$$\mathrm{d}Q_i \wedge \mathrm{d}P_i = \mathrm{d}q_i \wedge \mathrm{d}P_i + hm_i^{-1}\mathrm{d}P_i \wedge \mathrm{d}P_i,$$

but $\mathrm{d}u \wedge \mathrm{d}u \equiv 0$ for any u, hence

$$\mathrm{d}Q_i \wedge \mathrm{d}P_i = \mathrm{d}q_i \wedge \mathrm{d}P_i,$$

and (2.21) implies

$$\mathrm{d}Q_i \wedge \mathrm{d}P_i = \mathrm{d}q_i \wedge \mathrm{d}p_i - h\sum_{j=1}^{\mathrm{N_c}} \mathrm{d}q_i \wedge \frac{\partial^2 U}{\partial q_j \partial q_i}\mathrm{d}q_j.$$

It is then a simple exercise to show that

$$\sum_{i=1}^{\mathrm{N_c}}\sum_{j=1}^{\mathrm{N_c}} \mathrm{d}q_i \wedge \frac{\partial^2 U}{\partial q_j \partial q_i}\mathrm{d}q_j = 0,$$

using the skew-symmetry of the wedge product and the fact that the Hessian matrix is symmetric. This implies that

$$\sum_{i=1}^{\mathrm{N_c}} \mathrm{d}Q_i \wedge \mathrm{d}P_i = \sum_{i=1}^{\mathrm{N_c}} \mathrm{d}q_i \wedge \mathrm{d}p_i,$$

which means that the method is symplectic.

2.3.7 The Adjoint Method

Given any numerical integrator $\mathscr{G}_h$, consider the map

$$\mathscr{G}_h^{\dagger} = \mathscr{G}_{-h}^{-1}.$$